

OVERALL CONCEPT · 总体概念

Human-Robot Labor Confluence · 人机合辙
'Zhe' is the imprint a wheel leaves on the rail. CONFLUENT TRACKS translates the century-old collaborative spirit of Jing-Zhang switchmen — 'guarding the switch with their lives' — into a spatial institution of human-robot relay at the physical labor layer: robots and drones take on hazardous work such as power-line inspection, utility-culvert inspection and fire reconnaissance in authorized pilots, while humans retain a three-stage chain of control — authorization, halt and acceptance; AI algorithms assist testing of urban traffic green-wave coordination and transit priority.
Long-Life Frame + Fast AI Infill
'Replace the model, not the city' : a 50-year long-life spatial frame plus fast modular AI infill slots, exploring the lifecycle mismatch between rapidly iterating AI models and long-lived urban physical structures.
50-Year Frame / 6-Month Iteration
The spatial frame carries a 50-year service life while AI modules iterate on a ~6-month cadence; handover stations, twin loops and the land-use grid reserve unified physical interfaces for successive algorithm generations.

THREE LAWS OF CONFLUENCE · 合辙三律

I · Physical Labor Collaboration
Robots are proposed to take on elevated, confined and toxic operations in authorized test scenarios, replacing front-line workers' hazardous exposure.
II · Human Sovereign Final Review
Human pre-authorization, a retained mid-task physical e-stop, and post-task expert acceptance sign-off — the three-stage chain of control is never ceded.
III · Pedestrian Priority in Public Space
Low-speed robots and UAVs in mixed spaces take pedestrian and vulnerable-group priority as a proposed test objective.

Regional Synergy (Conceptual Research)

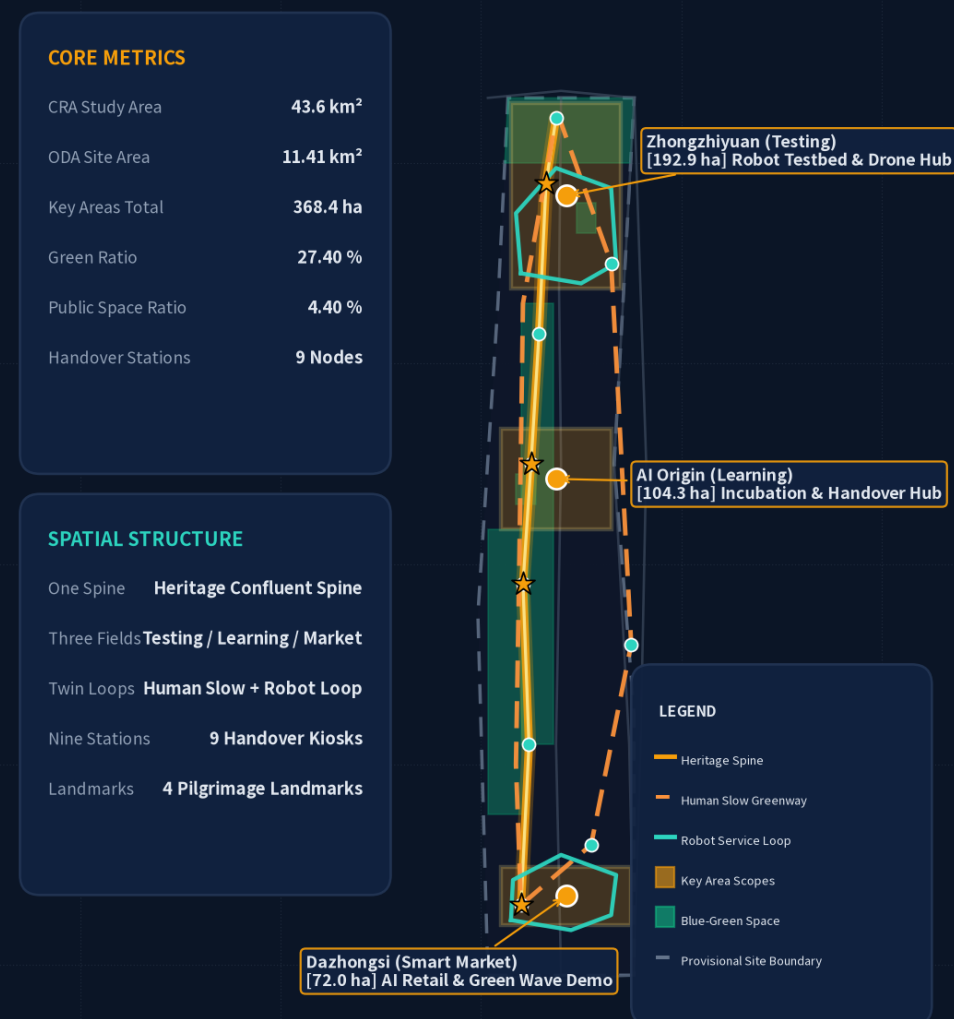
Haidian original innovation — Future Science City (energy AI) / Huairou Science City (fundamental compute) / Beijing E-Town (embodied manufacturing) regional transfer — Jing-Jin-Ji (Tianjin Port, Xiong'an, Shijiazhuang, Baoding) demonstration and compute support. This chain is a hypothesis for validation, not an existing collaboration, approved regional plan or funding arrangement.

NAMING & IDENTITY · 命名与品牌

合辙京张 / CONFLUENT TRACKS JING-ZHANG
Subtitles: Zhongzhiyuan 'Confluent Testing Arena' · AI Origin Community 'Confluent Learning Garden' · Dazhongsi 'Confluent Smart Market' ; two wings: Zhongguancun Technology Services Wing · Xiaoyue River Scenario Enablement Wing.
The logo direction is modeled on Zhan Tianyou's herringbone switch: two parallel arrows in humanist orange (workers) and robot cyan (agents) within a railway circle; purely geometric, with no real trademarks or licensed fonts. International theme: 'From Switchman to Co-Pilot' .



Brand cover: herringbone switch + paired human-robot arrows



The Jing-Zhang Heritage Park human-robot confluence vitality spine runs north-south, linking the three areas as the backbone of walking, cycling and green-space systems.

Confluence scenario fields of the three key areas – Testing Arena, Learning Garden and Smart Market – hosting differentiated testing, learning and consumption scenarios.

A human walking-and-cycling loop (greenway ≈ 8.7 km) and robot low-speed service loops (north + south, ≈ 8.2 km combined) carrying delivery, inspection and cleaning services.

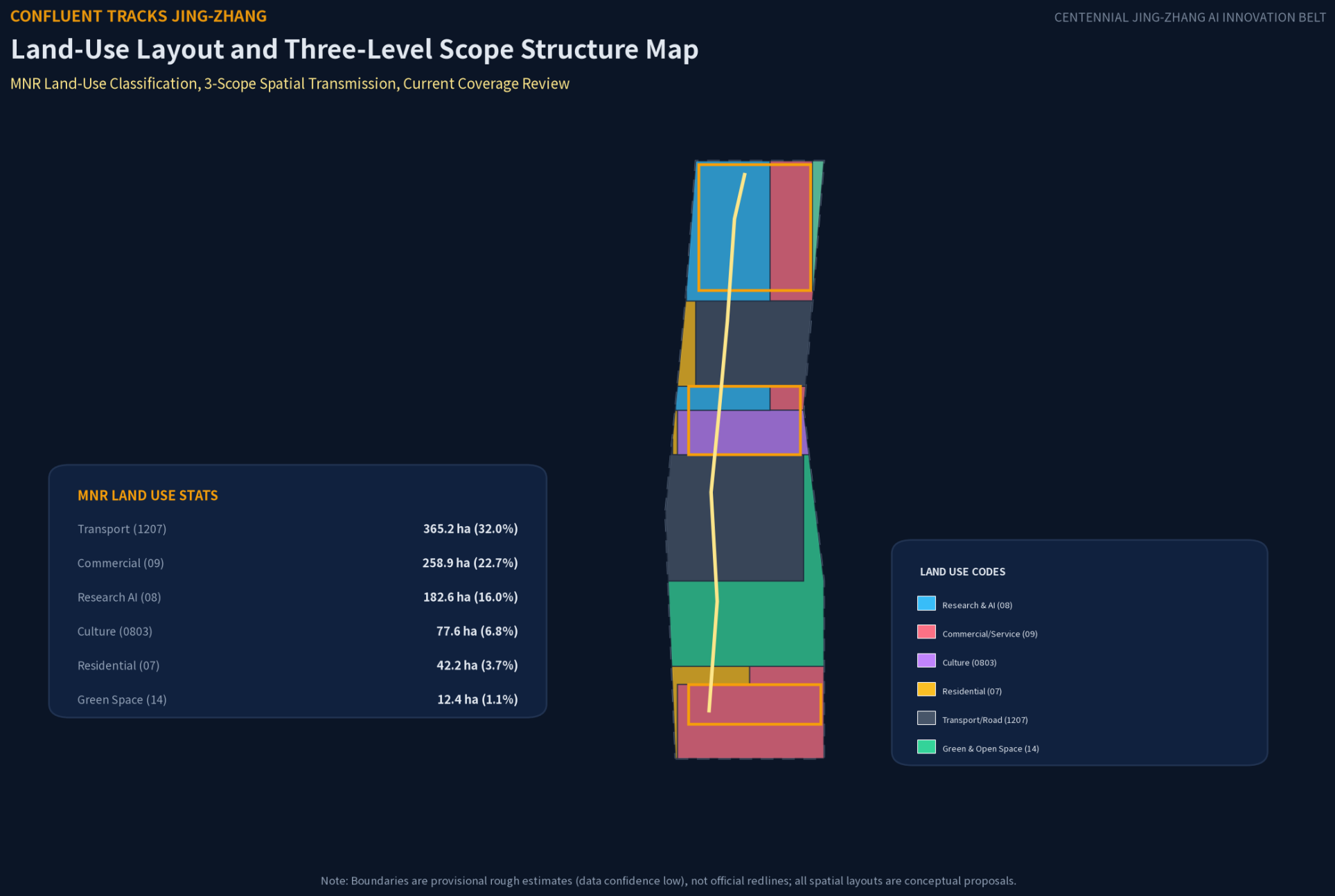
Nine human-robot handover stations — the spatial interface for pre-task authorization, mid-task halt and post-task acceptance — doubling as charging and data-return nodes.

Metric	Recalculated value	Basis
Overall design area	11,412,825.4 m ²	EPSC4548, +0.11% vs announcement
Green space / ratio	3,127,039.4 m ² / 27.4 %	5 green parcels
Public space / ratio	502,225.7 m ² / 4.4 %	6 core squares & landmarks
Conceptual footprint	3,710,756.5 m ² (54 bldgs)	conceptual volumes, not surveyed

Land use adopts topology-safe grid-clipped zoning with shared boundaries between adjacent polygons; current review leaves $\approx 100.8 \text{ m}^2$ uncovered inside the boundary and $\approx 0 \text{ m}^2$ outside; detailed relations await official redline review.

All spatial conclusions derive from a provisional rough boundary — not official redlines, statutory areas or ownership boundaries; the whole package is recalculated once official geometry is released.

Scope tier	Announcement	Recalculated (provisional)	Working role
Coordinated research	≈ 43.6 km ²	—	system research layer, three zones + two wings
Overall design area	≈ 11.4 km ²	11,412,825.4 m ² (+0.11%)	urban renewal & overall urban design
Key detailed areas	≈ 368.4 ha	192.9 / 104.3 / 72.0 ha	detailed design of three key areas



MNR 2023 LAND-USE RECALCULATION

MNR code	Category	Area (ha)	Share
07	Residential	42.03	3.7 %
08	Public admin & services	182.13	16.0 %
0803	Cultural (⊆08)	78.08	6.8 %
09	Commercial services	259.13	22.7 %
1207	Urban & rural roads	367.28	32.2 %
14	Green & open space	201.77	17.7 %
1401	Park green space (⊆14)	10.85	1.0 %

Note: 0803 ⊆ 08 and 1401 ⊆ 14 are never double-counted; 2023 MNR numeric codes are used (no ‘1301’ exists in the 2023 system — park green space is 1401).

URBAN RENEWAL FRAMEWORK · 更新框架

Low-Disruption Organic Renewal

Key renewal targets are Zhongzhiyuan potential land, near-campus blocks of the Origin Community, and the Dazhongsi commercial environment; AI agglomeration targets and industrial spatial scales are conceptual recommendations, and the total planned building volume remains pending confirmation without an official regulatory plan.

Retain-Renovate-Demolish · 拆改留

Retain heritage elements such as the Qinghuayuan Station historic site and quality existing buildings; renovate inefficient near-campus blocks and commercial environments; demolish only dilapidated structures confirmed by professional teams and valueless temporary buildings; build new mainly on potential land.

Conceptual Volumes · 概念体量

54 conceptual building volumes with a combined footprint of ≈ 3,710,756.5 m² — marked as conceptual-volume footprints, neither surveyed existing conditions nor approved scales.

DEVELOPMENT INTENSITY · 开发强度

The unknown / null Principle

Indicators depending on undisclosed official controls — floor area ratio, building height and building coverage — keep status=unknown, value=null, with recalculation paths documented in assumptions.json and metrics.json; conceptual volumes never masquerade as statutory control values.

Spatial Supply & Talent Life

R&D land sits close to rail transit and the walking/cycling loop; commercial-service land connects to robot service loops; residential land is placed in quieter western and southern areas; green and public spaces are evenly distributed to support innovation and exchange.

Architectural Character

Key areas balance future-oriented and cultural expression; controls on height, volume and roof form are recorded as conceptual guidance pending official conditions — heritage humility, new-area futurism and transition-zone coordination.

CONFLUENT TRACKS JING-ZHANG

Spatial Diagrams and Design Moves for Three Key Areas

Each area shows a simplified polygon, nodes, mobility loops and a risk marker

Zhongzhiyuan • Arena

192.9 ha • North 5th Ring

[Role]

Robot testbed / user AI lab service hub

[Landmark(s)]

Watchtower (stone wall + back)

[Risk]

Access, safety and QA review pending

AI Origin • Learning Garden

104.3 ha • Wudaokou Core

[Role]

AI incubation / on-governance demonstration

[Landmark(s)]

Handover Hub (infrastructure hub)

[Risk]

Heritage boundary and mooring pending

Dazhongsi • Smart Market

72.0 ha • Zhichunlu to Kailashan

[Role]

AI-native retail / modern services

[Landmark(s)]

Green Wave Crossing (four-quadrant walk)

[Risk]

Cyber and delivery robot right-of-way

Note: Key area boundaries are provisional, nodes, mobility and risks are conceptual diagrams, not official redlines or engineering alignments.

Detailed-design index of the three key areas

Unified Handover Interface

Before a task, authorization is confirmed by card or QR code; during a task it can be halted at any time in the halt zone; after a task, humans verify results in the acceptance zone. All three polygons are provisional and the designs are conceptual proposals for professional deepening.

Zhongzhiyuan • Confluent Testing Arena

≈ 192.9 ha (announcement reference 192.1 ha, comparison only) • North 5th Ring- Qing River

[Role] testing & validation field for inspection robots; proposed AI innovation testbed

[Landmark] Watchtower (drone nest + public viewing platform)

[Key projects] power-line / utility-culvert / bridge test grounds; Qing River waterfront garden

[Mobility] Fifth Ring Road area integration (conceptual, no redline conclusion)

[Risk] airspace and municipal authorizations pending; controls unknown

AI Origin Community • Confluent Learning Garden

≈ 104.3 ha • Tsinghua East Rd-Wudaokou

[Role] human-robot R&D and career-transition academy

[Landmark] Human-Robot Handover Hub (glazed visual authorization hall, S5 Wudaokou)

[Key projects] sci-tech incubation & transformation zones, healing gardens, launch functions

[Mobility] Wudaokou / Qinghua East Rd West station integration, campus-park slow links

[Risk] heritage boundary of Qinghuayuan Station site pending verification

Dazhongsi • Confluent Smart Market

≈ 72.0 ha • Zhichunlu-Xizhimen North

[Role] AI-native consumption and business cluster

[Landmark] Green-Wave Intersection (four-quadrant walking connection)

[Key projects] agents / smart terminals / content formats; delivery-robot narrow-lane microcirculation

[Mobility] metro linkage & greenway stitching; static traffic completed

[Risk] courier vs delivery-robot right-of-way requires field testing

Beiwei Community • Living Interface (Conceptual Research)

The residential living layer of the AI Origin Community: a community-level micro handover station (Type-C variant), an AI Living Lab unit and a community data dashboard; smart service stations, AI science corners, elderly-greeting and guide-companion routes. All content is conceptual exploration, not an existing collaboration or approved renewal.

12 Scenario Cards (3 Industrial Validation)

#	AI Scenario Card	Location	SWB Fallback
1	Power-Line Inspection Robot Partner (validation)	Zhongzhiyuan substation & line corridor	manual verification & breaker pull
2	Utility-Culvert Inspection Drone (validation)	Zhongzhiyuan utility tunnel	manual ventilation & manual entry
3	Traffic Green-Wave Coordination AI (validation)	Dazhongsi four-quadrant intersection	fixed timing + on-site police control
4	Bridge Structural Health Inspection	Qing River bridges & overpasses	manual basket inspection
5	Fire-Reconnaissance Vanguard Drone	belt-wide low-altitude grid corridors	traditional scout team entry
6	Night Quiet Cleaning Robot	twin loops & heritage park greenways	daytime manual sanitation
7	Guide-Cane Companion Robot	heritage park barrier-free greenway	white cane + volunteer guide
8	Commercial Micro-Delivery Robot	Dazhongsi dedicated narrow lanes	courier direct delivery
9	Emergency Medical AED Drone	belt-wide emergency low-altitude corridor	authorized ground-ambulance backup
10	Construction Safety AI Supervision	key-area renewal sites	conventional manual inspection
11	Autonomous Feeder Shuttle	station-integrated feeder routes	standard human-driven feeder bus
12	Solitary-Elderly Greeting Robot	Origin Community & senior zones	routine social-worker visits

Each card specifies data categories, anonymization, retention, access, human review, physical e-stop and complaint handling; response times, hotlines and access channels are proposed SLAs / test targets pending authorization — not current service commitments.

Three Spatial Conflict Resolution Prototypes

① Intersection Crosswalk Confluence A 1.5 m dedicated robot deceleration island and dynamic ground projection alerts; pedestrians hold the highest priority; robots decelerate to 2 km/h at 2 m and yield statically — validated against mixed-crossing collision-risk reduction.
② Courier vs Delivery-Robot Lane Separation A 1.2 m delivery-robot lane in Dazhongsi with a 15 cm sloped landscaped curb as physical buffering — tested against sideswipe-risk reduction between 25 km/h courier scooters and 6 km/h delivery robots.
③ Tactile Blind-Path Buffer A continuous 60 cm physical buffer on both sides of tactile paving along the 8.7 km greenway, barring charging kiosks or parked equipment; guide-companion robots hold a 50 cm lateral offset via UWB geofencing.

User Personas · 用户画像

Persona	Core need
Electrician Mr. Zhang (45)	fewer high-voltage climbs; robot partners take hazards
Traffic assistant Ms. Li	smarter signals, but she can take over
Engineer Xiao Wang (25)	a walkable ‘third space’
Heritage docent Mr. Chen	heritage respected; tactile, readable narrative
Visually impaired Ms. Liu	tactile & audible guidance; unobstructed blind paths
Courier Xiao Zhou	safe cycling space not crowded out by robots

AI Pilgrimage Landmarks · 朝圣地标

4 Landmarks (all conceptual) ① Centennial Switch Square (south Zhongzhiyuan) — switchman sculpture beside an inspection robot with a tactile relief timeline; ② Human-Robot Handover Hub (S5 Wudaokou) — visual authorization hall; ③ Watchtower (Qing River) — drone nest + public viewing platform; ④ Green-Wave Intersection (Dazhongsi) — data-flow lighting visualizes signal coordination at night.

CONFLUENT TRACKS JING-ZHANG

Mobility and Blue-Green Public Space Integrated System Map

Twin Mobility Loops, Heritage Park Spine, Continuous Ecology & AI Hotspots

CENTENNIAL JING-ZHANG AI INNOVATION BELT

MOBILITY NETWORK

Total Road Length

50.6 km

Slow Mobility Net

16.9 km

Human Greenway Loop

8.7 km

Robot Service Loops

8.2 km

Handover Stations

9 Nodes

BLUE-GREEN ECOLOGY

Green Space Area

312.7 ha

Green Ratio

27.46 %

Public Space Area

50.2 ha

Public Space Ratio

4.46 %

Water Corridors

Qinghe / Xiaoyue

Note: Boundaries are processed rough estimates (data confidence low), not official redlines; all spatial layouts are conceptual proposals.

Composite mobility & blue-green public-space system

Blue-Green Metrics · 蓝绿指标

Green space ≈ 312.7 ha / 27.4 %; public space ≈ 50.2 ha / 4.4 %. The heritage-park vitality belt is the north-south spine; the Qing and Xiaoyue rivers form east-west blue-green corridors with two pocket parks.

Twin Loops & Roads · 双环与道路

System	Recalc. length	Note
Road network	50,618.2 m (≈ 50.6 km)	existing centerlines are schematic, not redlines
Human slow loop	greenway ≈ 8.7 km	along the heritage park & eastern links
Robot service loops	north + south ≈ 8.2 km	delivery / inspection / cleaning
Slow + service total	16,898.8 m (≈ 16.9 km)	conceptual, unvalidated by simulation

AI Traffic Optimization

At the Dazhongsì four-quadrant demonstration segment, AI dynamically coordinates signals with transit priority; traffic police retain physical override, and every automated decision can be human-reviewed and halted. Feeder connections link Wudaokou, Qinghua East Road West Entrance, Dazhongsì and Xizhimen stations.

Nine Handover Stations · 九处交接站

Type	Stations	Footprint	Functions
Type-A Hub	S5 Wudaokou	≈ 1,200 m²	visual authorization hall · dispatch center · workshop
Type-B District	S1 / S2 / S4 / S8	≈ 300-500 m²	charging berths · anonymization pods · duty rooms
Type-C Outpost	S3 / S6 / S7 / S9	≈ 80-120 m²	battery swap · shelter · emergency supplies

Three-Tier Wayfinding (distinct from the belt logo)

Tier-1 regional wayfinding at main entrances, zone boundaries and transit connections; Tier-2 district wayfinding at station forecourts and greenway junctions; Tier-3 site identification for facilities, accessibility and heritage nodes. Braille strips, raised plans and NFC-triggered bilingual audio-guide posts follow Article 39 of the Barrier-Free Environment Construction Law.

Values recalculated from provisional rough geometry, not official redlines · All spatial and operational proposals are conceptual · metrics traceable to metrics.json

Board 6 / 7

CONFLUENT TRACKS JING-ZHANG

Core Metrics Recalculation and Evidence Dashboard

EPSC 4548 Recalculated, 4 self-check records, Honest Boundary Disclosure

Total Site Area (ODA)

11,412,825.4 m²

2242 km² / Brief 2140 km² (45.11%)

Green Ratio (green_ratio)

27.40 %

Green Area 3,127,039.4 m² (EPSC 4548)

Public Space Ratio

4.40 %

Public Area 502,225.7 m² (4% Roadside)

1. Raw GeoJSON

Vector Layers (WGS84)
Current Boundary/Contour Review

→

2. Projection

EPSC 4548 Projected
Under other Projection

→

3. Recalculation

33 Metrics Verified
Written to metrics.json

→

4. Self-Check Gate

Spatial/Visual Evidence
Current package checks

Verified Known Metrics Table (EPSC 4548)

Site Area (ODA)	11,412,825.4 m ²
Green Area / Ratio	3,127,039.4 m ² (27.40%)
Public Space / Ratio	502,225.7 m ² (4.40%)
Building Footprint	3,710,756.5 m ² (34 Blocks)
Road Network Length	50,618.2 m (26.6 km)
Slow Mobility Length	15,610.8 m (16.9 km)

Data Gaps and Compliance Disclaimers

A-BOUNDARY-001	Provisional rough boundary / Pending official redline
A-CONTROLS-001	FAB/Height remain unknown / No fabricated controls
A-HERITAGE-001	Heritage elements subject to official relic records
A-THEFTS-001	Road lines schematic / Green wave 8 conceptual
A-PHASING-001	Phasing is conceptual / No municipal fiscal commitment
A-STANDARDS-001	MOHURD arch standard marked data_gap pending fix

Note: Boundaries are provisional rough estimates (data confidence low), not official redlines; all spatial layouts are conceptual proposals.

Core metrics recalculation & evidence chain

Metrics & Evidence · 指标体系

Metric	Status	Value
site_area_sqm	known	11,412,825.4 m²
green_ratio	known	27.4 % (3,127,039.4 m²)
public_space_ratio	known	4.4 % (502,225.7 m²)
road_network_length_m	known	50,618.2 m
slow_service_network_length_m	known	16,898.8 m
buildings / footprint	known	54 bldgs / 3,710,756.5 m²
floor_area_ratio	unknown	null (pending official plan)
building_height_m_max	unknown	null (pending official plan)
building_density_official	unknown	null (pending official plan)

The compliance matrix covers Announcement 1.3/1.4/1.5 and agent.1-agent.6; 9 professional standards (all 5 mandatory addressed); all 15 design-depth items complete; all four self-check gates PASS.

Policy & Operation · 政策与运营

On-Duty Voucher & Access Timetable

Hazardous inspection robots and autonomous shuttles entering public space pass closed-track validation and receive electronic on-duty vouchers before authorization; nighttime low-noise (22:00-06:00) and daytime off-peak schedules may be tested. All policy and funding arrangements are conceptual recommendations.

Long-Term Operation & Annual Congress

An annual Human-Robot Confluence Congress, a developer challenge on hazardous-operation simulation tracks, a scenario-access channel for enterprises, public experience routes and conversion monitoring — under the international theme ‘From Switchman to Co-Pilot’ .

Three Risk Categories · 风险

① Official redlines and key-area polygons not yet obtained — all spatial conclusions are provisional with a package-wide recalculation trigger; ② regulatory controls, road redlines, ownership and municipal pipelines are not public — related metrics remain unknown; ③ heritage scopes and facts (Qinghuayuan Station site, Juesheng Temple) await verification by cultural-relic authorities.

Implementation Phasing · 实施分期

Phase	Period	Area	Focus
Near-term pilots	2026-2028	≈ 159.4 ha	three station pilots + green-wave segment
Mid-term network	2028-2030	≈ 297.4 ha	twin loops connected; nine stations formed
Long-term confluence	2030-2035	≈ 392.3 ha	43.6 km² synergy & long-term operation

All phasing is a conceptual recommendation, not a development-timeline commitment or government approval.